



بسم الله الرحمن الرحيم

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Design of PID Controller for controlling fluid temperature of Heat Exchanger System

A thesis Submitted As Partial Fulfillment of Requirements of the
Degree of Master in Control Engineering

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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

الآية

﴿وَأَنْزَلَ اللَّهُ عَلَيْكَ الْكِتَابَ وَالْحِكْمَةَ وَعَلَّمَكَ مَا لَمْ تَكُنْ تَعْلَمُ وَكَانَ فَضْلُ اللَّهِ عَلَيْكَ عَظِيمًا﴾

صدق الله العظيم

[سورة النساء: 113]

Dedication

Specially dedicated to my mother, father, and those people who have guided and inspired me through this work.

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Foremost praise is to Allah, the almighty. Second, I would like to express Appreciation and thanks to my adviser **Dr. Ate Abd Elrahim** for the continuous support of my M.Sc study and research. I am forever grateful to the M.sc College of neelain University the College which gave me the opportunity to study M.Sc in control engineering. My appreciation also goes to my family who has support me all these years thanks for their encouragement, Love emotional support that they had given to me.

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List of Abbreviations

Tubular Exchanger Manufacturers Association	TEMA
India Standard	IS
Proportional-Integral-Derivative	PID
Settling time	S-T
Rising time	R-T
Over shoot	O-S
Peak amplitude	P-A
Final value	F-V
Steady state error	SS-E
Feedback	F-B
Feed Forward	F-F
Mile mper	mA
Kilo gram	Kg
pascal	Psig
Set point	X_{sp}
Measured value	$X_m(t)$
Rate of out let temperature	$\Delta\theta$
Rate of steam flow	Δm

Abstract

Heat exchanger system is widely used in chemical plants because it can sustain wide range of temperature and pressure. The main purpose of a heat exchanger system is to transfer heat from a hot fluid to a cooler fluid, so temperature control of outlet fluid is of prime importance. Because any limit increasing or decreasing in the temperature may causes some problems in the stability of the system. Any Increased in process temperature may be reaching the Ignition point such as in Petrochemical Industries. The proposed approach employs a PID feedback controller and feedback plus feed-forward controller to regulate the temperature of an outlet fluid of a shell and tube heat exchanger to a certain reference value. The transient performance and the error criteria of the controllers are analyzed and the best controller is found out. From the simulation results, it is found out that the PID controller has best the performance comparison with feedback plus feed forward controller. The modal is developed using Simulink/MATLAB and simulations results are generated.

المستخلص

تستخدم المبادلات الحرارية على نطاق واسع في العمليات الصناعية والكيميائية وذلك لأنه بإمكانها المحافظة على درجة الحرارة والضغط على نطاق واسع . الغرض الرئيسي من أنظمة المبادلات الحرارية هو عملية نقل الحرارة من سائل ساخن إلى سائل آخر بارد دون الخلط بينهما لذلك يعتبر التحكم في درجة حرارة السائل الخارج ذو أهمية قصوى , أي زيادة أو نقصان في درجة الحرارة عن الحد المسموح به يمكن أن يسبب بعض المشاكل في استقرار النظام , كما في بعض التطبيقات الصناعية مثل الصناعات البترولية فإن زيادة درجة الحرارة قد توصل السائل الى نقطة الاشتعال .

يستخدم النظام المقترح في هذا البحث المتحكممة التناسبية التكاملية التفاضلية للحلقة المغلقة بالإضافة للمسار الامامي وذلك لتنظيم درجة حرارة السائل الخارج من المبادل الحراري في حدود نقطة مرجعية معينة . كما تم تحليل استجابة اداء النظام ومعايير الخطاء وتم الحصول على افضل تحليلات ونتائج عند استخدام المتحكممة التناسبية التكاملية التفاضلية , مقارنة بتحليلات المسار الامامي . تمت عملية المحاكاة وتحليلات الاداء باستخدام برنامج الماتلاب .

Chapter One

Introduction

1.1 General View

Heat exchangers are devices that are used to transfer enthalpy between two or more fluids, between a solid surface and a fluid, or between solid particulates and a fluid, at different temperatures. There are different types of heat exchanger used in the industry but most of the industry use shell and tube type heat exchanger. It consists of parallel tubes enclosed in a shell. There is a variety of application of heat exchanger system. Some of the applications include electronic cooling, refrigeration and air conditioning, manufacturing, and power generation. [6]

In This research reports a work that considers a heat exchanger and builds a single-input single outputs model of the system with the help of experimental data available. The outlet temperature of the heat exchanger system has to be kept at a desired set point according to a process requirement. [1]

1.2 Problem Statement

Temperature control of the shell and tube heat exchanger is characteristics of nonlinear, time varying and time lag. In Heat exchangers increase or decrease in temperature for the limit be causes some problem in stability of the system.

For example there are some applications for heat exchangers any Increased in process temperature may be reach the Ignition point as such as in Petrochemical Industries.

1.3 Research Objectives

The main objectives of this research are:

To improve the response of the fluid temperature of heat exchanger system, (settling time, overshoot) to maintain the temperature of the heat exchanger at a certain value according to the needs of the process.

1.4 previous Studies

Masoud Abedinifar and Sarhad Azizian were designed intelligent controller by combination between fuzzy and PID controller to control of shell and tube heat exchangers. Then they founded desired temperature has a shortest possible time and minimum overshoot irrespective of step change in load and process disturbances [1].

Yuvraj Bhushan Khare and Yaduvir Singh were took a case study of heat exchanger system, and then they evaluate different methods to control the outlet fluid temperature. Four different kinds of controllers were designed to control the outlet temperature of fluid and the performances of these controllers were evaluated by two different methods. One of the methods they used for performance evaluation was the time domain analysis of overshoot and settling time and the other method was calculation of performance Indices.

Firstly a classical PID controller was designed to achieve the control objective. But due to the unsatisfactory performance of the PID controller a feed forward controller then were designed and placed in the forward path of the system. To further increase the efficiency of the system the internal model based PID controller was designed and implemented.

The internal model based PID controller gives satisfactory performance in both steady state and transient state in time domain analysis [2].

Navid Khalili Dizaji, Aidin Sakhvati and Seyed Hossein Hosseini were designed a PID feed-forward controller for controlling output fluid temperature in shell and tube heat exchanger. Maximum mutation in this control system was 9.1% and settlement time is 90. 23 seconds, which in chemical process's settlement time about few minutes. [3]

Subhransu Padhee implemented different controller (feedback, feedback plus feed-forward and internal model controller) to control the outlet temperature of a shell and tube heat exchanger system. Mathematical model of the heat exchanger was developed using experimental data and the process model is used to develop the respective controller. The performance of different controllers was evaluated using transient characteristics and error indices. From the simulation results, she is found that the internal model control has a superior performance than feedback and feedback plus feed-forward controller.

The feedback controller implemented using classical PID controller shows a higher degree of overshoot and settling time whereas the internal model control negates the overshoot and has a manageable settling time. [4]

Soniya Kocher and Dr. A.K. Kori were designed controller for controlling the Temperature Heat Exchanger process of a plant .and they studied effects of the controller on heat exchanger is analyzed .

Automatic tuning of the controller is also Introduce for efficient control of the process .and they tuned is implemented using Ziegler Nichols method and they compared results with Variation in temperature without controller [5].

1.5 Methodology

This research will be focus about designing and tuning a PID controller with many tuning rules where the main objective is to form such a tuning rule which can be characterized from the mathematical model of the system. The three parameters of PID controller are tuned using auto tuning method. The PID controller was runs with feed-forward controller in order to reach the control purpose; also analyzing designed controller performance is done in the time domain.

In order to evaluate the designed controller performance, unit step response analysis and performance indicators were used. Processes modeling and controllers' design indicators steps have implemented in MATLAB / Simulink.

1.6 Thesis Layouts

This research consists of five chapters. **Chapter one** is an introduction which is about modeling. **Chapter two** is a literature review about heat exchanger and mathematical modeling of the system. **Chapter three** describes the control strategies and design procedures of a PID controller. **Chapter four** is a Results and discussion for results. **Chapter five** are given Conclusion and recommendations.

Chapter Two

Literature review about heat exchanger

2.1 Introduction

A heat exchanger is a tool this is used to transfer thermal energy (enthalpy) among two or greater fluids, among a solid floor and a fluid, or among stable particulates and a fluid, at different temperatures and in thermal touch. In heat exchangers, there are generally no outside heat and paintings interactions. Traditional packages involve heating or cooling of a fluid movement of concern and evaporation or condensation of unmarried- or multi component fluid streams. In different programs, the goal may be to get better or reject heat, or sterilize, pasteurize, fractionate, distill, concentrate, crystallize, or control a manner fluid. In a few heat exchangers, the fluids exchanging warmth are in direct touch. In most heat exchangers, heat switch among fluids takes place through a keeping apart wall or into and out of a wall in a brief way. In lots of heat exchangers, the fluids are separated by a heat switch floor, and preferably they do no longer blend or leak. Such exchangers are referred to as direct switch kind , or simply recuperates.[2]

In comparison, exchangers in which there is intermittent warmth exchange between the hot and cold fluids through thermal energy storage and launch via the exchanger surface or matrix are called indirect transfer kind, or regenerators. Such exchangers typically have fluid leakage from one fluid circulate to the other, because of strain differences and matrix rotation/valve switching. Common examples of warmth exchangers are shell-and tube exchangers, car radiators, condensers, evaporators, air pre heaters, and cooling towers. If no phase change occurs in any of the fluids in the exchanger, it is sometimes referred to as a sensible heat exchanger. There could be internal thermal energy sources in the exchangers, such as in electric heaters and nuclear fuel elements. [6]

2.2 Classification of heat exchangers

Transfer of heat from one fluid to every other is an important operation for most of the chemical industries. The most commonplace application of heat transfer is in designing of heat transfer device for exchanging heat from one fluid to any other fluid. Such gadgets for efficient switch of heat are typically known as heat Exchanger .heat exchangers are typically categorized depending on the switch manner occurring in them. General classification of heat exchangers is shown in the Figure 2.1. [7]

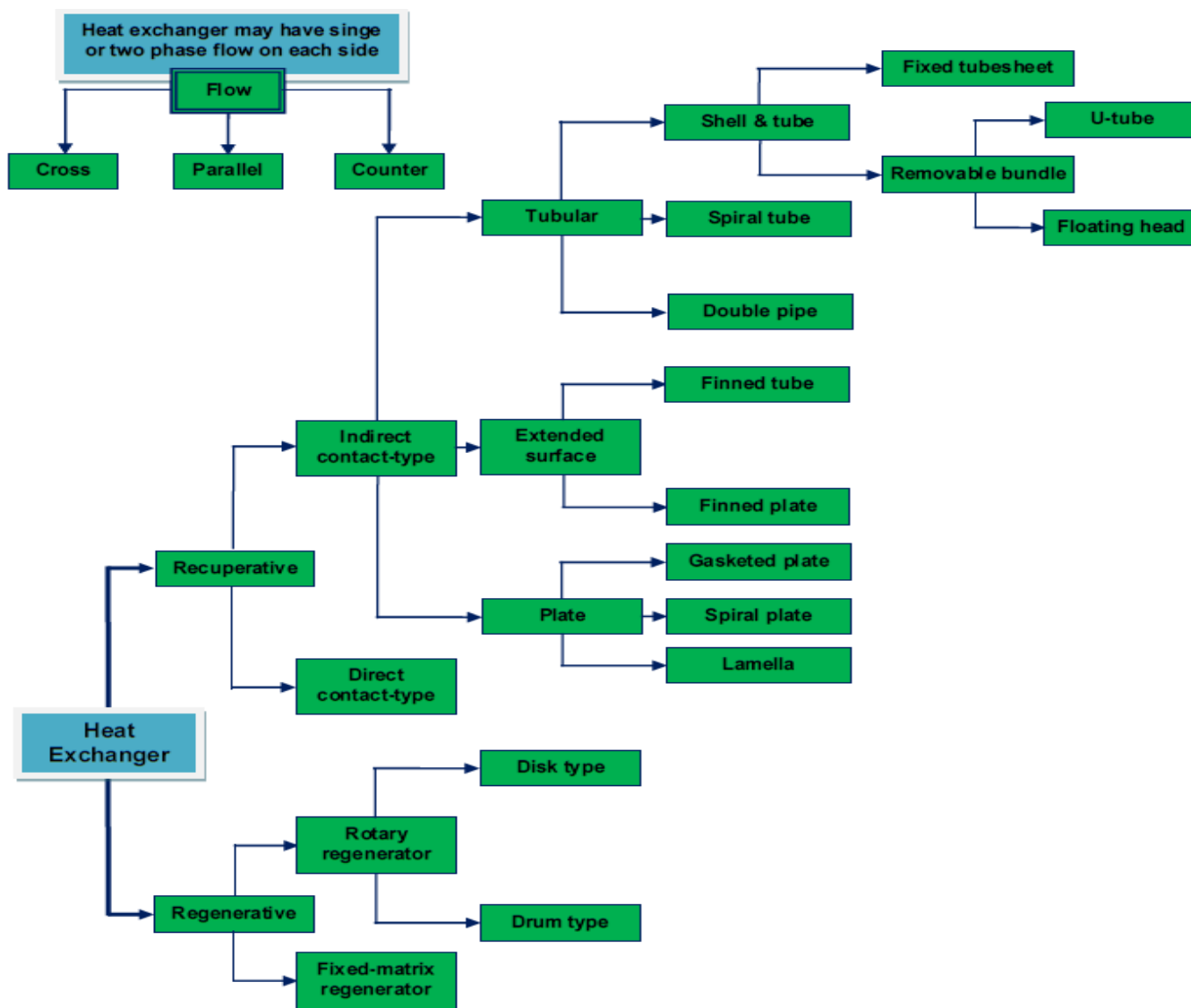


Figure: 2.1 Classification of heat exchangers depending on their applications.

Among of all sort of exchangers, shell and tube exchangers are most generally used heat exchange device. The common kinds of shell and tube exchangers are:

2.2.1 Fixed tube-sheet exchanger (non-removable tube bundle)

The best and most inexpensive type of shell and tube exchanger is with constant tube sheet design. On this form of exchangers the tube sheet is welded to the shell and no relative motion between the shell and tube package deal is possible Figure2.2. [7]

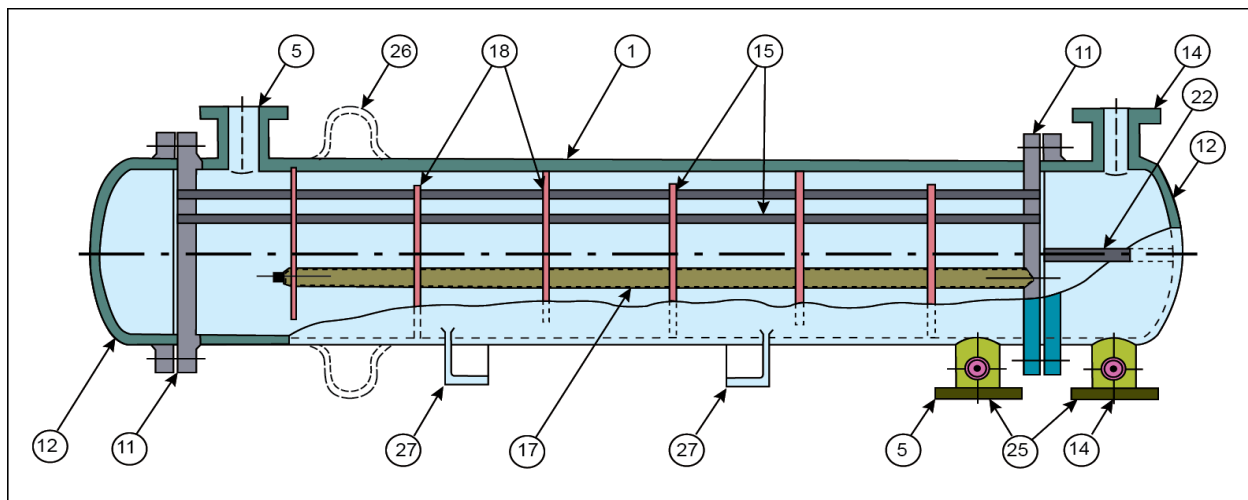


Figure: 2.2 Fixed-tube heat exchanger

2.2.2 Removable tube bundle

Tube package may be removed for ease of cleansing and replacement. Removable tube package deal exchangers similarly can be categorized in floating-head and U-tube exchanger.

Floating-head exchanger: It includes of a stationary tube sheet that is clamped with the shell flange. At the alternative quit of the package, the tubes may also amplify right into a freely using floating-head or floating tube sheet.

A floating head cover is bolted to the tube sheet and the whole package deal may be removed for cleaning and inspection of the indoors. This kind of exchange is shown in Figure 2.3.

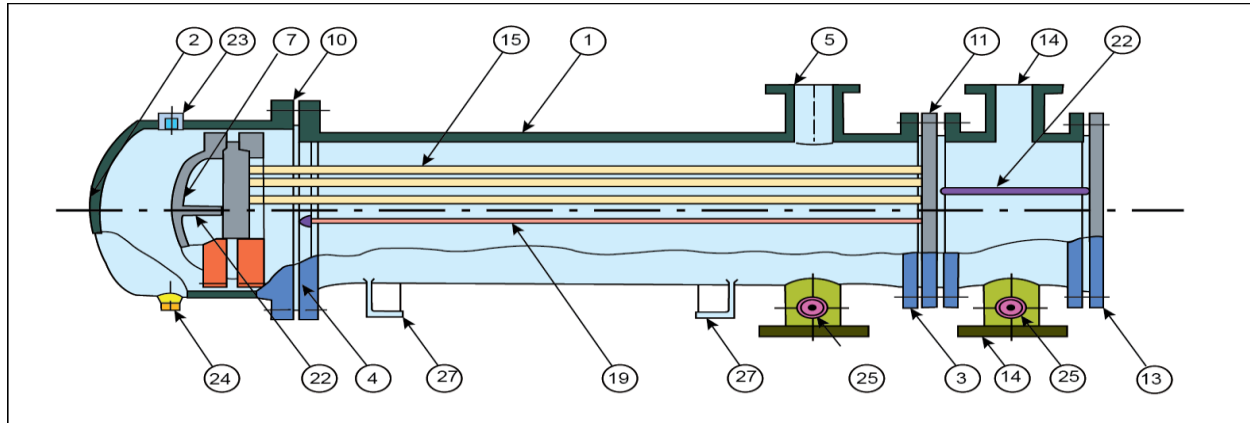


Figure: 2.3 Floating-head heat exchanger (non-pull through type)

U-tube exchanger: This kind of exchangers consists of tubes that are bent in the form of a „U“ and rolled again into the tube sheet shown inside the Figure 2.4. This means that it's going to remove some tubes at the center of the tube bundle depending on the tube association. The tubes can extend freely closer to the „U“ bend end [7].

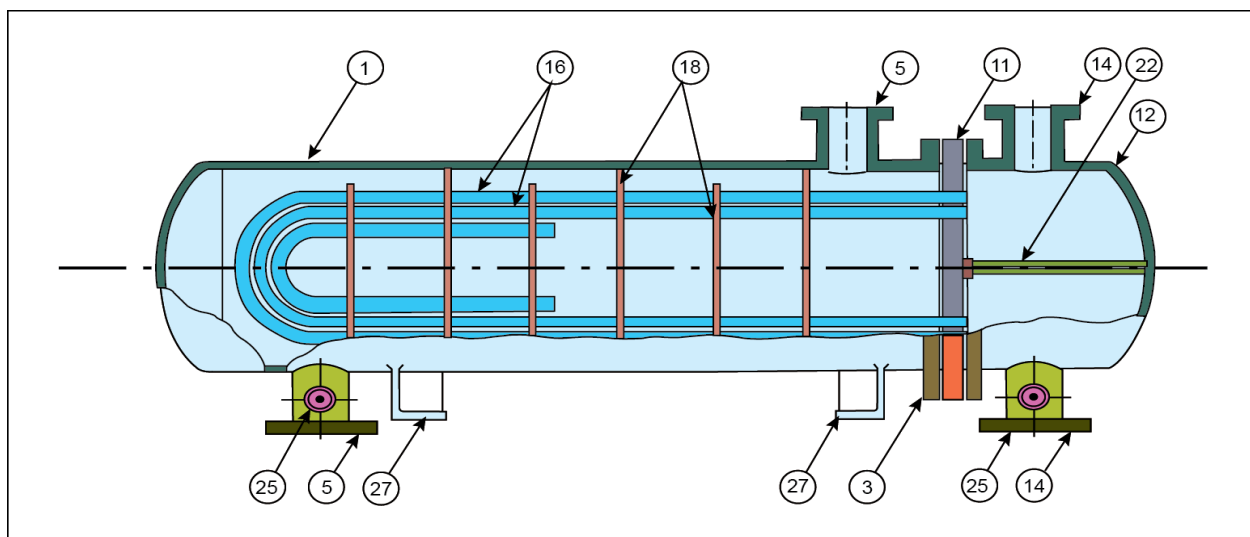


Figure: 2.4 Removable U-tube heat exchanger

Typical parts and connections shown in Figures 2.2, 2.3 and 2.4 (IS: 4503-1967) are summarized below.

- | | |
|--------------------------------|---|
| 1. Shell | 16. Tubes (U-type) |
| 2. Shell cover | 17. Tie rods and spacers |
| 3. Shell flange (channel end) | 18. Transverse (or cross) baffles or support plates |
| 4. Shell flange (cover end) | 19. Longitudinal baffles |
| 5. Shell nozzle or branch | 20. Impingement baffles |
| 6. Floating tube sheet | 21. Floating head support |
| 7. Floating head cover | 22. Pass partition |
| 8. Floating head flange | 23. Vent connection |
| 9. Floating head gland | 24. Drain connection |
| 10. Floating head backing ring | 25. Instrument connection |
| 11. Stationary tube sheet | 26. Expansion bellows |
| 12. Channel or stationary head | 27. Support saddles |
| 13. Channel cover | 28. Lifting lugs |
| 14. Channel nozzle or branch | 29. Weir |
| 15. Tube (straight) | 30. Liquid level connection |

The specific operational and constructional blessings and limitations depending on packages of shell and tube exchangers are summarized in Table 2.1. TEMA (United States) and IS: 4503-1967 (India) requirements provide the suggestions for the mechanical design of unfired shell and tube heat exchangers. As proven inside the Table 2.1, TEMA 3-digit codes specify the kinds of the front-stop, shell, and rear-end of shell and tube exchangers [7].

Table 2.1 TEMA 3-digit codes

Shell and Tube Exchangers	Typical TEMA code	Advantages	Limitations
Fixed tube sheet	BEM, AEM, NEN	Provides maximum heat transfer area for a given shell and tube diameter. Provides for single and multiple tube passes to assure proper velocity. Less costly than removable bundle designs.	Shell side / outside of the tubes are inaccessible for mechanical cleaning. No provision to allow for differential thermal expansion developed between the tube and the shell side. This can be taken care by providing expansion joint on the shell side.
Floating-head	AEW, BEW, BEP, AEP, AES, BES	Floating tube sheet allows for differential thermal expansion between the shell and the tube bundle. Both the tube bundle and the shell side can be inspected and cleaned mechanically.	To provide the floating-head cover it is necessary to bolt it to the tube sheet. The bolt circle requires the use of space where it would be possible to place a large number of tubes. Tubes cannot expand independently so that huge thermal shock applications should be avoided. Packing materials produce limits on design pressure and temperature.
U-tube	BEU, AEU	U-tube design allows for differential thermal expansion between the shell and the tube bundle as well as for individual tubes. Both the tube bundle and the shell side can be inspected and cleaned mechanically. Less costly than floating head or packed floating head	Because of U-bend some tubes are omitted at the centre of the tube bundle. Because of U-bend, tubes can be cleaned only by chemical methods. Due to U-tube nesting, individual tube is difficult to replace. No single tube pass or true countercurrent flow is possible. Tube wall thickness at the U-bend is

		designs.	thinner than at straight portion of the tubes. Draining of tube circuit is difficult when positioned with the vertical position with the head side upward.
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A typical chemical process for heating consists of a chemical reactor and a shell and tube heat exchanger system. The heat exchanger heats up the cold fluid to a desired point using super-heated steam supplied from the boiler. The storage tank provides cold fluid in 32°C using a pump and one-way valve. The super-heated steam comes from the boiler and flows through the shell side, whereas, the cold fluid flows through the tube side of the shell and tube heat exchanger system show Figure 2.5.

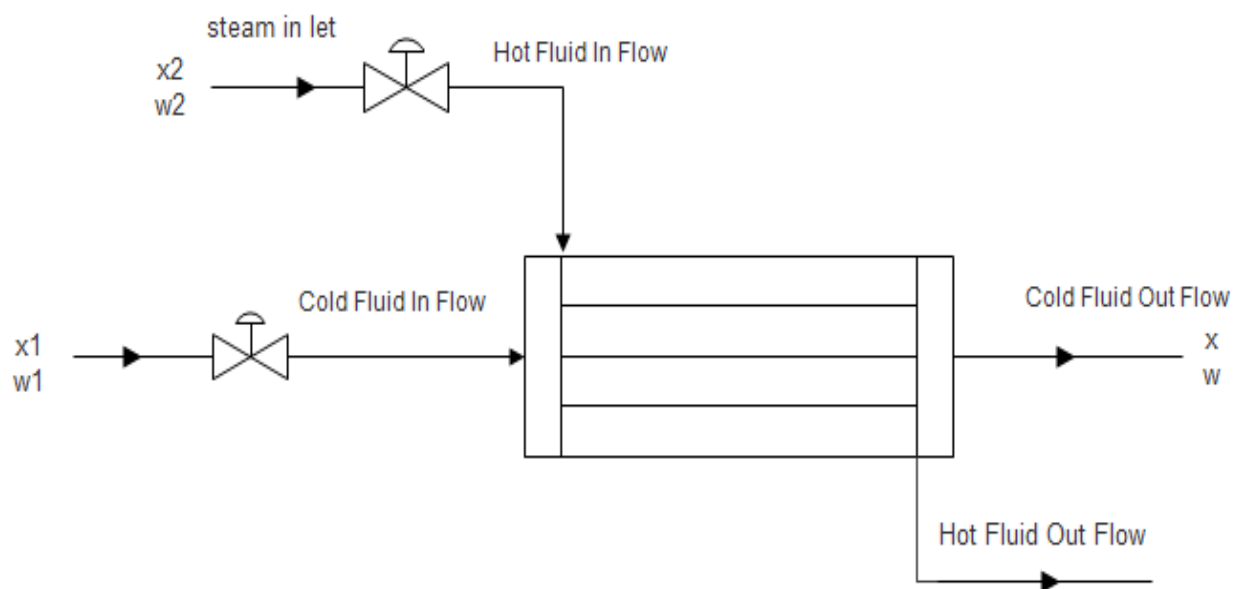


Figure: 2.5 show basic principle of heat exchanger

After the steam heats up the cold fluid, the condensed steam at 93°C goes out of the steam trap. Consequently, the heat exchanger heats up the cold fluid to a temperature of 52°C. The control objective is to control the temperature of the outlet cold fluid.[3]

Different assumptions have been considered in present work. The first assumption is that the inflow and the outflow rate of fluid are same. The second is that the heat storage capacity of the insulating wall is negligible. Figure 2.6 shows the feedback control scheme adopted in heat exchanger, thermocouple, sensing element, is implemented in the Feedback path of the control architecture. The temperature of the outgoing fluid is measured by the thermocouple and the output of the thermocouple is sent to the transmitter unit, which eventually converts the thermocouple output to a standardized signal in the range of 4-20mA. Output of the transmitter unit is given to the controller unit. The controller implements the control algorithm, compares the output with the set point and then gives necessary command to the final control element via the actuator unit [1].

The actuator unit is a current to pressure converter and the final control unit is an air to open valve. The actuator unit takes the controller output in the range of 4-20mA and converts it in to a standardized pressure signal in the range of 3-15 psig. The valve actuates according to the controller decisions.

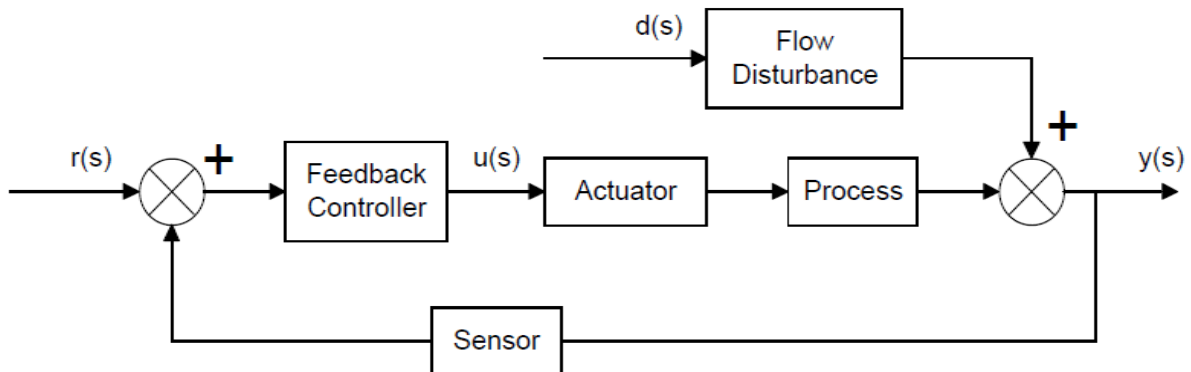


Figure: 2.6 Shell and tube heat exchanger's feedback control scheme

There are two types of disturbances in this process, one is the flow variation of input fluid, and the second is the temperature variation of input fluid which has significant effect compared to first one[1].

2.3 Feedback PID Control Theory

A proportional–integral–derivative controller (PID controller) is a control loop feedback mechanism. As the name suggests, PID algorithm consists of three basic coefficients: proportional, integral and derivative which are varied to get optimal response.

The entire idea of this algorithm revolves around manipulating the error. The error as is evident is the difference between the Process Variable and the Set point.

$$\text{ERROR} = \text{PV} - \text{SP}$$

These three modes are used in different combinations:

P – Sometimes used

PI - Most often used

PID – Sometimes used

PD – Very rare, useful for controlling servomotors

The ideal version of the PID controller is given by the formula

$$u_{(t)} = K_P * e_{(t)} + k_i * \int_0^t e_{(\tau)} d\tau + k_d \frac{de}{dt} \quad 2.1$$

Where u is the control signal and e is the control error ($e = r - y$). The reference value, r, is also called the set point.

The control signal is thus a sum of three terms: a proportional term that is proportional to the error, an integral term that is proportional to the integral of the error, and a derivative term that is proportional to the derivative of the error.[13]

The controller parameters are proportional gain k_p , integral gain k_i and derivative gain k_d . The controller can also be parameterized as

$$u(t) = K_P \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right) \quad 2.2$$

The main adjusting parameters of the PID controller are:

K_P = constant of proportionality, which describes P component T_i = integral time, which describes I component, ($K_i = 1/T_i$) T_d = derivative time, which describes D component, ($K_P = T_d$) If different values of the adjusting parameters are chosen, two variations of PID graphs are obtained as shown in the following graphs.

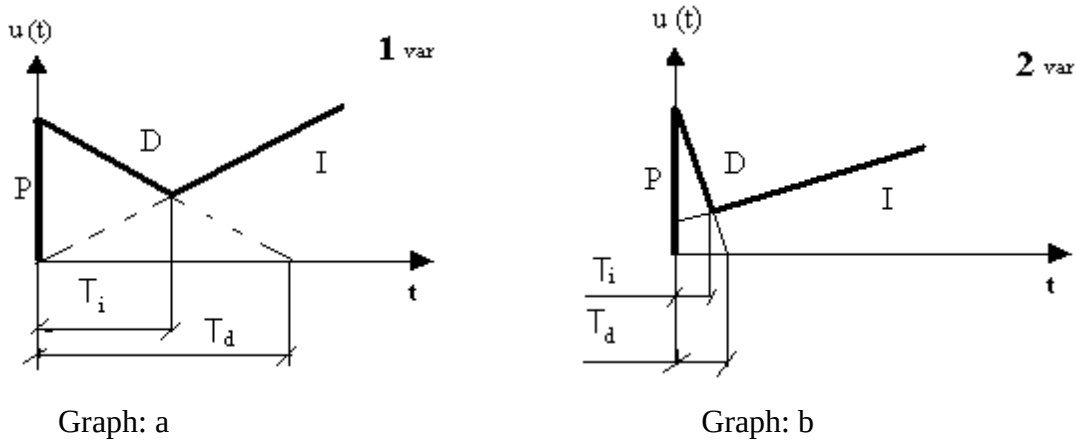


Figure: 2.7 Variation of PID control mode [12]

From Figure: 2.7 graph (a) it is clear that when different values are chosen to control the parameter, the derivative time (T_d) which describes the D component and the integral time which describe the I component overshoot but on the other hand when other values are chosen for graph (b) the derivative time and the integral time are shortened.

That means that it is very important for the operator to know the kind of values to be chosen that best suits his operation to make the system stable at all stages.

2.4 Effects of Proportional, Integral and Derivative Action

The four main characteristics of a closed-loop step responds that make the system Unstable are as follows:

1. Rise Time the time taken for the plant output to rise above 90% of the desired level for the first time.
2. Overshoot how much the peak level is higher than the steady state, normalized against the steady state.
3. Settle Time the time it takes for the system to converge to its steady state.
4. Steady-state Error the difference between the steady-state output and the desired output.

These characteristics make the system unstable and using the *PID* control mode to control the parameters such as k_p , k_i , k_d , the system will be stable.

The effects of increasing each of the controller parameters k_p , k_i , k_d can be summarized as Table 3.1 below

Table: 2.2 effects of increasing each of the controller parameters

Response	Rise Time	Over shoot	Settling Time	S-S Error
k_p	Decrease	Increase	NT	Decrease
k_i	Decrease	Increase	Increase	Eliminate
k_d	NT	Decrease	Decrease	NT

NT: No definite trend Minor change.

k_p is used to decrease the rise time while the k_d is used to reduce the overshoot and the setting time. k_i is used to eliminate the steady-state error[10].

2.5 PID Tuning Methods

Tuning is adjustment of control parameters to the optimum values for the desired control response. Stability is a basic requirement. However, different systems have different behavior, different applications have different requirements, and requirements may conflict with one another. PID tuning is a difficult problem, even though there are only three parameters and in principle is simple to describe, because it must satisfy complex criteria within the limitations of PID control. There are accordingly various methods for loop tuning, some of them:

- ✓ Manual tuning method
- ✓ Ziegler–Nichols tuning method
- ✓ PID tuning software methods

2.5.1 Manual Tuning Method:

In manual tuning method, parameters are adjusted by watching system responses. K_p , K_i , K_d are changed until desired or required system response is obtained. Although this method is simple, it should be used by experienced personal. One Manual Tuning Method Example: Firstly, K_i and K_d are set to zero. Then, the K_p is increased until the output of the loop oscillates, after obtaining optimum K_p value, it should be set to approximately half of that value for a "quarter amplitude decay" type response. Then, K_i is increased until any offset is corrected in sufficient time for the process. However, too much K_i will cause instability.

Finally, K_d is increased, until the loop is acceptably quick to reach its reference after a load disturbance. However, too much K_d also will cause excessive response and overshoot. A fast PID loop tuning usually overshoots slightly to reach the set point more quickly; however, some systems cannot accept overshoot, in which case an over-damped closed-loop system is required, which will require a K_p setting significantly less than half that of the K_p setting causing oscillation. In Table 3.1, the effects of changing control parameters can be seen.[13]

2.5.2 Ziegler Nichols Tuning Method

Ziegler and Nichols performed a series of experiments to determine the values of k_p , k_i and k_d based upon transient step response of a plant. The two were able to arrive with two methods which were simply named the first and second methods. First method which applied a step response and it is typical of first order system with transportation delay. The response gives an S-curve sometimes known as reaction curve when drawn and it is characterized by two parameters, L the delay time and T the time constant which is illustrated in Figure 2.8, this time is determined by drawing a tangent to the step response at its point of inflection and noting its intersection with the time axis and the steady-state value [11].

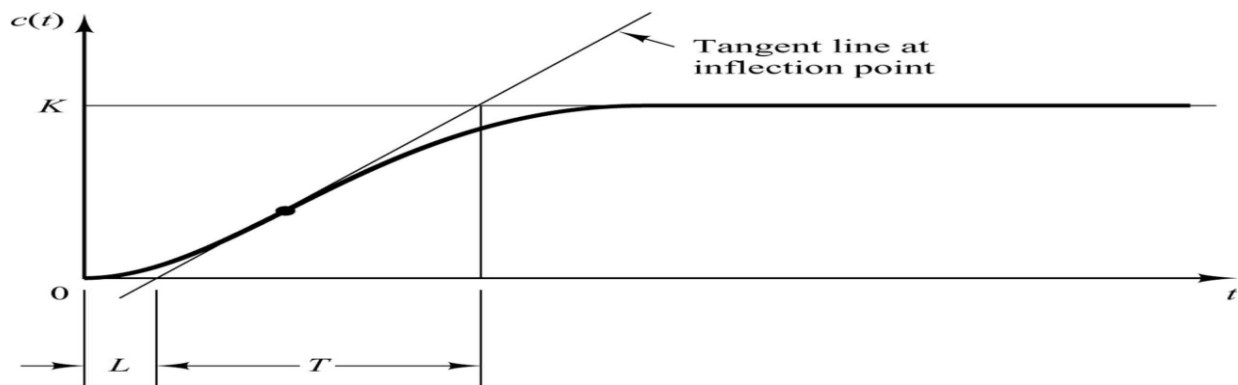


Figure: 2.8 Response Curve for Ziegler-Nichols First Method

Using the L and T parameters as explained above, K_p , K_i and K_d can be set according to Ziegler-Nichols Model as shown in Table 2.3.

Table: 2.3 Ziegler-Nichols Models - First Method

PID Type	k_p	$T_i = k_p/k_i$	$T_d = k_d/k_p$
P	T/L	∞	0
PI	$0.9 T/L$	$L/0.3$	0
PID	$1.2 T/L$	$2 L$	$0.5 L$

Second Method the continuous cycling method this is a closed-loop technique whereby, using proportional control only, the controller gain K_1 is increased until the system controlled output $c(t)$ oscillates continually at constant amplitude, like a second-order system with no damping.

This condition is referred to as marginal stability, this value of controller gain is called the ultimate gain k_u , and time period for one oscillation of $c(t)$ is called the ultimate period T_u show Figure 2.9 below.

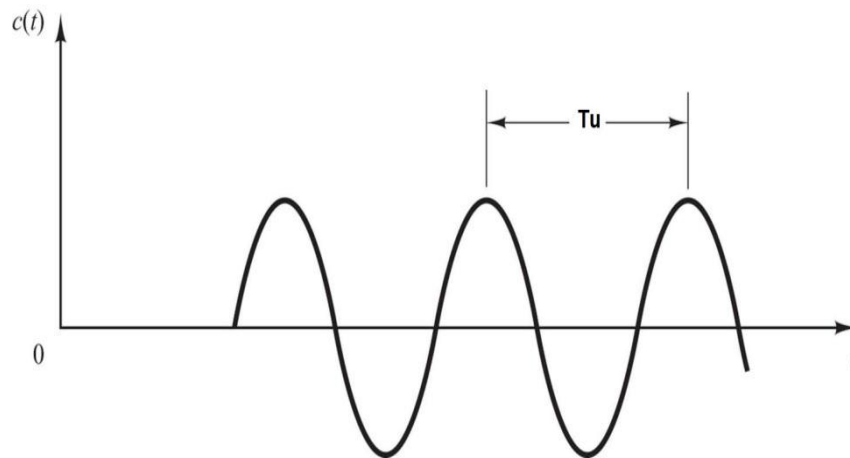


Figure: 2.9 ultimate periods T_u

The controller parameters, as a function of k_u and T_u , to provide a similar closed-loop response to the process reaction method, are given in Table 2.4.

Table: 2.4 Ziegler- Nichols PID parameters using the continuous cycling method

Type of controller	K1	T_i	T_d
P	$0.5 k_u$	∞	0
PI	$0.45 k_u$	$0.833 T_u$	0
PID	$0.6 k_u$	$0.5 T_u$	$0.125 T_u$

2.5.3 PID Tuning Software

There is some prepared software that they can easily calculate the gain parameter. Any kind of theoretical methods can be selected in some these methods. Some Examples:

- MATLAB Simulink PID Controller Tuning,
- BESTune, Exper Tune etc

2.5.3 Cohen-Coon Method

In this method the process response curve is obtained first, by an open loop test as shown in figure8 and then the process dynamics is approximated by a first order plus dead time model, with following parameters:

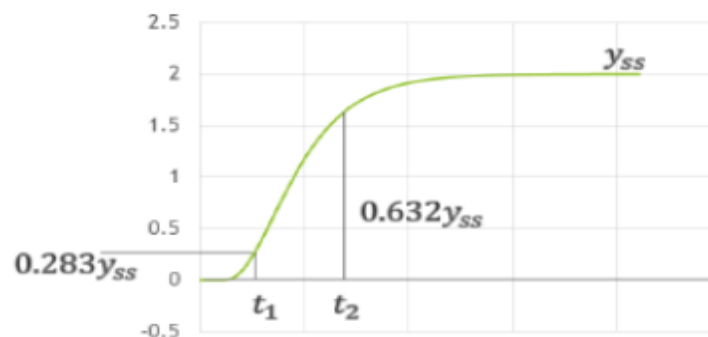


Figure 2.10: The open loop response of plant

Again the particular rules for this method are used to calculate the PID parameters. They are listed in Table 2.5 and Table 2.6 shown overview of tuning methods

$$\tau_m = \frac{3}{2}(t_2 - t_1)$$

$$t_d = t_2 - \tau_m$$

2.3

Table 2.5: the parameter of Cohen-Coon Method

Controller	K_c	T_l	T_d
P	$\frac{\tau_m}{K t_d} \left(1 + \frac{t_d}{3\tau_m} \right)$	-	-
PI	$\frac{\tau_m}{K t_d} \left(0.9 + \frac{t_d}{12\tau_m} \right)$	$t_d \left(\frac{30 + \frac{3t_d}{\tau_m}}{9 + \frac{20t_d}{\tau_m}} \right)$	-
PD	$\frac{\tau_m}{K t_d} \left(1.25 + \frac{t_d}{6\tau_m} \right)$	-	$t_d \left(\frac{6 - \frac{2t_d}{\tau_m}}{22 + \frac{3t_d}{\tau_m}} \right)$
PID	$\frac{\tau_m}{K t_d} \left(1.33 + \frac{t_d}{4\tau_m} \right)$	$t_d \left(\frac{32 + \frac{6t_d}{\tau_m}}{13 + \frac{8t_d}{\tau_m}} \right)$	$t_d \left(\frac{4}{11 + \frac{2t_d}{\tau_m}} \right)$

Table 2.6 overview of tuning methods

Method	Advantages	Disadvantages
Manual Tuning	No math required. Online method	Requires experienced personnel
Ziegler-Nichols	Proven Method. Online method	Process upset, some trial –and error, very aggressive tuning
Software tools	Consistent tuning .Online or Offline method .May includes valve and sensor analysis .Allow simulation before downloading. Can support non-steady State (NSS) Tuning	Some cost and training involved
Cohen-Coon Method	Good process models	Some math, offline method ,only good for first –order processes

2.6 Mathematical Modeling

In industrial process heat exchanger is used to transfer heat from a hot fluid through a solid wall to a cooler fluid. In the proposed model shell and tube type exchanger is used, which is very adaptable, flexible and can operate over the full range of pressures and temperatures encountered in chemical processes. [3]

They are generally composed of a bundle of tubes which are contained in a single jacket called “shell”. One of the fluids flows in the tubes and the other flows inside the shell around the tubes. Its construction is simple and economical, but the outside surface of the tubes is not accessible for mechanically cleaning. The mathematical model gives the transfer function which reveal the terms that describe the process response: gain, Time Constant, Dead Time.

In the heat exchanger the fluid that flows through the inner pipe at constant velocity ‘ v ’ is heated by steam condensing outside the pipe. The temperature of the fluid entering the pipe and the steam temperature vary according to some arbitrary functions of time. The steam temperature varies with them, but not with position in the exchanger. The metal wall separating steam from fluid is assumed to have significant thermal capacity that must be accounted for in the analysis.

The heat transfer from the steam to the fluid depends on the heat-transfer coefficient on the steam side (h_o) and the convective transfer coefficient on the water side (h_i). The resistance of the metal wall is neglected. The goal of the analysis is to find transfer functions relating the exiting fluid temperature $T(L,t)$ to the entering fluid temperature $T(0,t)$ and the steam temperature $T_v(t)$. [8].

Heat Exchanger transfer function $G_p(s)$ relating the outlet temperature D_q to the steam flow D_m and a time constant of 30 sec.

$$G_p(s) = \frac{\Delta\theta/\Delta m}{\tau s + 1} = \frac{105/2.1}{30s + 1} = \frac{50}{30s + 1} \quad 2.4$$

2.7 Control Valve Dynamics

Control valve implements the decision of the controller. They receive the signal from the controller and accordingly adjust the value of manipulated variable. Commonly used final control element is the air-operated valve, which controls the flow through an orifice by positioning the plug appropriately. [8]

The pneumatic control valve is the most commonly used final control element. In this case Air to close type pneumatic and an equal percentage valve is used.

For this valve, the position of the stem (or, equivalently, of the plug at the end of the stem) will determine the size of the opening for flow and consequently the size of the flow (flow rate). The position of the stem is determined by the balance of all forces acting on it. These forces are:

$P_d A$ = Force exerted by the compressed air at the top of the diaphragm.

Kx = Force exerted by the spring attached to the stem and the diaphragm.

$C \frac{dx}{dt}$ = Frictional force exerted upward and resulting from the close contact of the stem with valve packing.

Where:

A – Area of the Diaphragm, P_d – Pressure acting on diaphragm, x – Displacement, K – Hook's constant, C – friction coefficient between stem and packing.

The dynamics of a pneumatic valve can be approximated by that of second order system as

$$\frac{x(s)}{p_d(s)} = \frac{A/K}{\tau s + 1} \quad 2.5$$

Diameter of diaphragm = 273 mm now evaluation of K: At steady state, $P_d A = Kx$; rherefore

$$K = \frac{p_d A}{x} \quad 2.6$$

$A = 90.73 \text{ inch}^2$

$K = 15 * 90.73 / 1.56 = 872.4 \text{ psi/mm}$;

Here $x = 1.56 \text{ mm}$ & $P_d = 15 \text{ psi}$.

Control Valve Transfer Function The control valve has a maximum travel of 1.56 mm, linear characteristics and a time constant of 3 sec. The nominal pressure range of the valve is 3 to 15 psig.

Control Valve Gain

$$= \frac{\text{Range of stem}}{\text{pressure range}}$$

$$= \frac{1.56 \text{ mm}}{(3 - 15)\text{psi}}$$

$$= 0.13 \text{ psi/mA}$$

The total transfer-function of Actuator

$$G_V(s) = \frac{0.13}{3s + 1} \quad 2.7$$

Sensor Transfer Function:

The sensor has a calibrated range of 00 to 2000C and a time-constant of 1 to 10sec.
Sensor gain is

$$= \frac{(20 - 4)mA}{(100 - 0)^{\circ}C} = 0.16 \text{ mA}/^{\circ}C$$

The transfer-function of the sensor $H(s)$

$$H(s) = \frac{0.16}{10s + 1} \quad 2.8$$

For mathematical modeling, the heat exchanger system, actuator, valve, sensor are modeled using the available experimental data. The experimental process data is summarized in Table2.7 [1]. Figure 2.11 shown Shell and tube heat exchanger system control scheme.

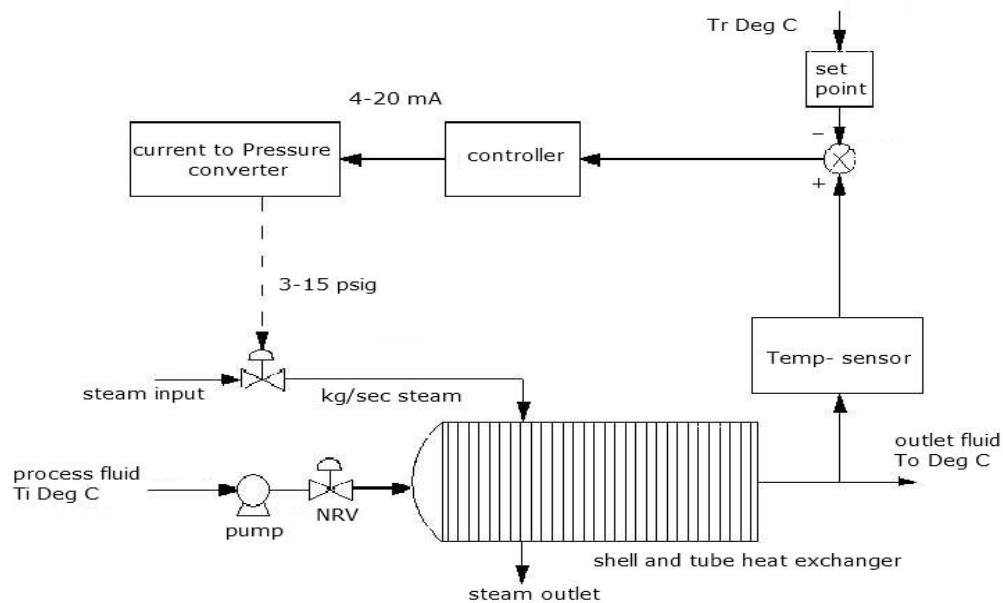


Figure: 2.11 Shell and tube heat exchanger system control scheme

Table 2.7 Experimental data of heat exchanger control process

Heat exchanger response to the steam flow gain	$50^{\circ}\text{C}/(\text{kg}/\text{sec})$
Time constants	30 sec
The current range of actuator	4-20 mA
The pressure range of actuator	3-15 psig
Control valve capacity	1.6 kg /sec
Time constant of control valve	3 sec
The range of temperature sensor	50-150 $^{\circ}\text{C}$
Time constant of temperature sensor	10 sec

The transfer functions of different part of the feedback control of shell and tube heat exchanger system and related gains are shown in Table 2.8 [1].

Table 2.8 Transfer functions and gains

Process transfer function	$\frac{50e^{(-s)}}{30s + 1}$
Transfer function of valve	$\frac{0.13}{3s + 1}$
Gain of current to pressure convertor	0.75
Transfer function of disturbance	$\frac{1}{30s + 1}$
Transfer function of temperature sensor	$\frac{0.16}{10s + 1}$

In this control system, feed-forward controller regulates distortion by the input fluid stream. Feed-forward transformation function is shown in $G_{cf}(s)$, which is expressed as equation 2.8 [3]:

$$G_{cf}(s) = -\frac{G_d(s)}{G_p(s)} \quad 2.9$$

In equation 2.8, $G_p(s)$ is a process transformation function and $G_d(s)$ is a distortion transformation function for the input stream, which are shown in equations 2.10 and 2.11, respectively.

$$G_p(s) = \frac{5e^{-s}}{90s^2 + 33s + 1} \quad 2.10$$

$$G_d(s) = \frac{1}{30s + 1} \quad 2.11$$

By substituting current distortion transformation function and process transformation function in equation 2.9, feed-forward controlling function is expressed as equation 2.12. [8]

$$G_{cf}(s) = \frac{-18s^2 - 6.6s - 0.2}{27s^2 + 30.9s + 1} \quad 2.12$$

Using experimental data, transformation functions and rates in shell and tube heat exchanger control system are shown in Figure 2.12: [3]

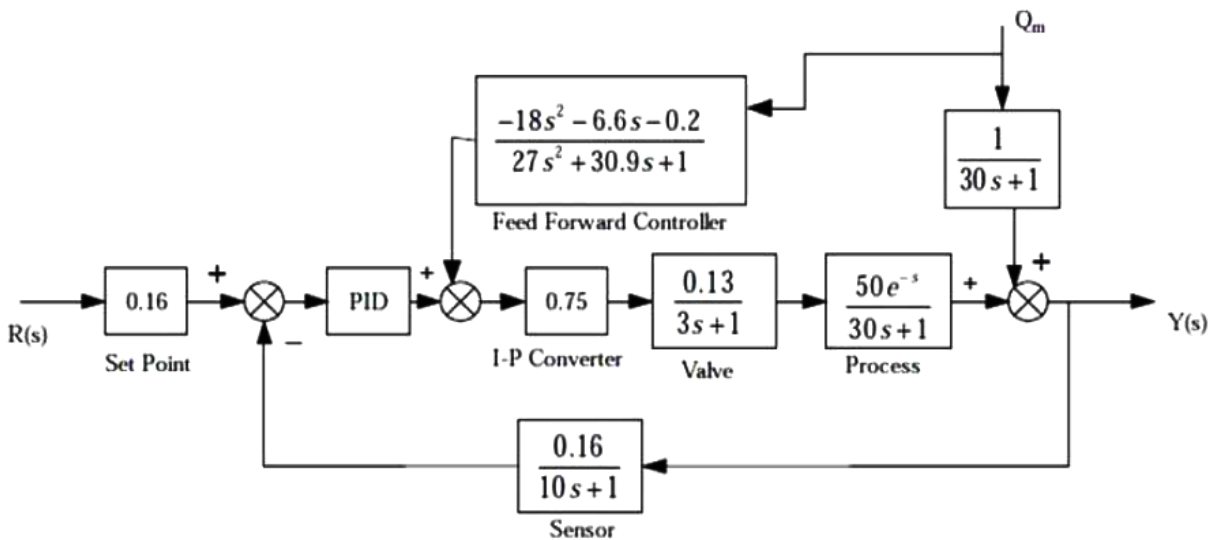


Figure: 2.12 Transformation functions and rates in sell and tube heat exchanger control system

Chapter Three

Control system design of the heat exchanger

3.1 Control Strategies

In this research several strategies for controlling output fluid temperature in shell and tube heat exchanger are used as shown in the following

3.1.1 Strategy No. 1

Measure x and adjust w_2 . It is reasonable to measure controlled variable x and then adjust w_2 accordingly. For example, if x is too high, w_2 should be reduced; if x is too low, w_2 should be increased. This control strategy could be implemented by a person (manual control). However, it would normally be more Convenient and economical to automate this simple task (automatic control), this Strategy called feedback method. [9]

X is Measured Variable and w_2 is Manipulated Variable. A schematic diagram of strategy1 is shown in Figure 3.1. The outlet temperature is measured and transmitted to the controller as an electrical signal. (Electrical signals are shown as dashed lines in Figure 3.1. The controller executes the control law and sends the calculated value of w_2 to the control valve as an electrical signal. The control valve opens or closes accordingly.

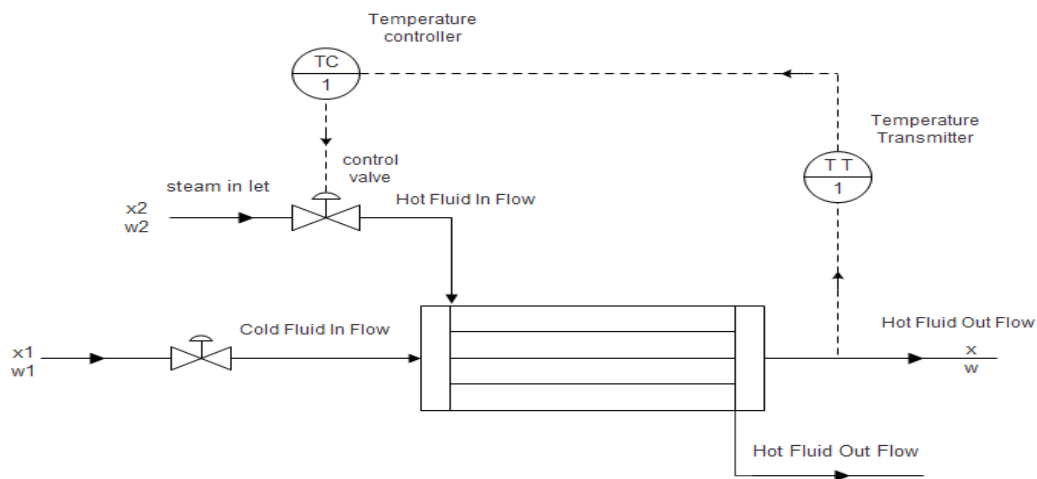


Figure: 3.1 A schematic diagram of strategy1 (feedback)

Consider the equipment that is used to implement control strategies. For heat exchanger system under feedback control in Figure 3.1, the exit temperature x is controlled and the flow rate w_2 of pure steam is adjusted using proportional control. To consider how this feedback control strategy could be implemented, a block diagram for Heat exchanger control system is shown in Figure 3.2.

Operation of the temperature control system can be summarized for the key hardware components as follows:

Analyzer and transmitter: The temperature of heat exchanger exit is measured by means of an instrument that generates a corresponding mille ampere (mA)-level signal. This time-varying signal is then sent to the controller.

Feedback controller: The controller performs three distinct calculations. First, it converts the actual set point X_{sp} into an equivalent internal signal X_{sp}' . Second, it calculates an error signal $e(t)$ by subtracting the measured value $X_m(t)$ from the set point X_{sp}' that is, $e(t) = X_{sp}' - X_m(t)$.

Control valve: The controller output $p(t)$ in this case is a DC current signal that is sent to the control valve to adjust the valve stem position, which in turn affects flow rate $w_2(t)$. Because many control valves are pneumatic, i.e., are operated by air pressure, the controller output signal may have to be converted to an equivalent air pressure signal capable of adjusting the valve position.

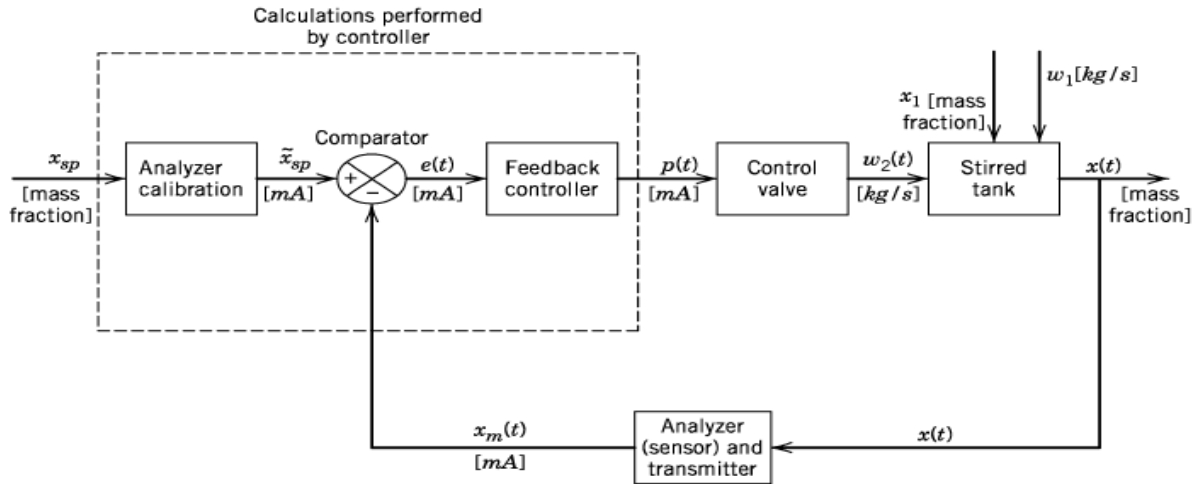


Figure: 3.2 Block diagram for composition feedback control system

The block diagrams in Figure 3.2 provide a convenient starting point for analyzing process control problems. The physical units for each input and output signal are also shown.

The schematic diagram in Figure 3.1 Shows the physical connections between the components of the control system, while the block diagram shows the flow of information within the control system. The block labeled "control valve" has $p(t)$ as its input signal and $w_2(t)$ as its output signal, which illustrates that the signals on a block diagram can represent either a physical variable such as $w_2(t)$ or an instrument signal such as $p(t)$. Each component in Figure 4 exhibits behavior that can be described by a differential or algebraic equation [9].

3.1.2 Strategy NO. 2

Measure x_1 , adjust w_2 . As an alternative to Method 1, we could measure disturbance variable x_1 and adjust w_2 accordingly. Figure 3.3 below shown schematic diagram of strategy 2.

This method called feed forward strategy because we measured disturbance signal (cold fluid in flow) to controlling for steam in let flow (hot fluid in flow). [9]

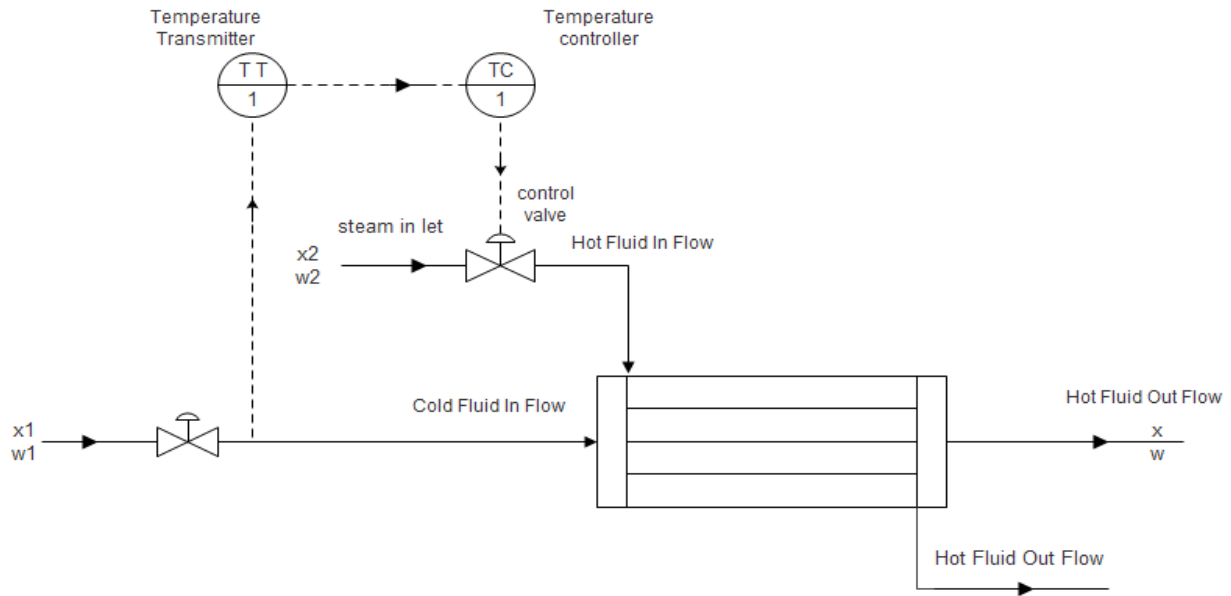


Figure: 3.3 A schematic diagram of strategy2 (feedforward)

3.1.3 Strategy NO.3

Measure x_1 and x , adjust w_2 . This approach is a combination of strategy1 and 2 this method a combined feedforward-feedback control strategy because both x and x_1 are measured. Figure 3.4 below shown schematic diagram of strategy3 [9].

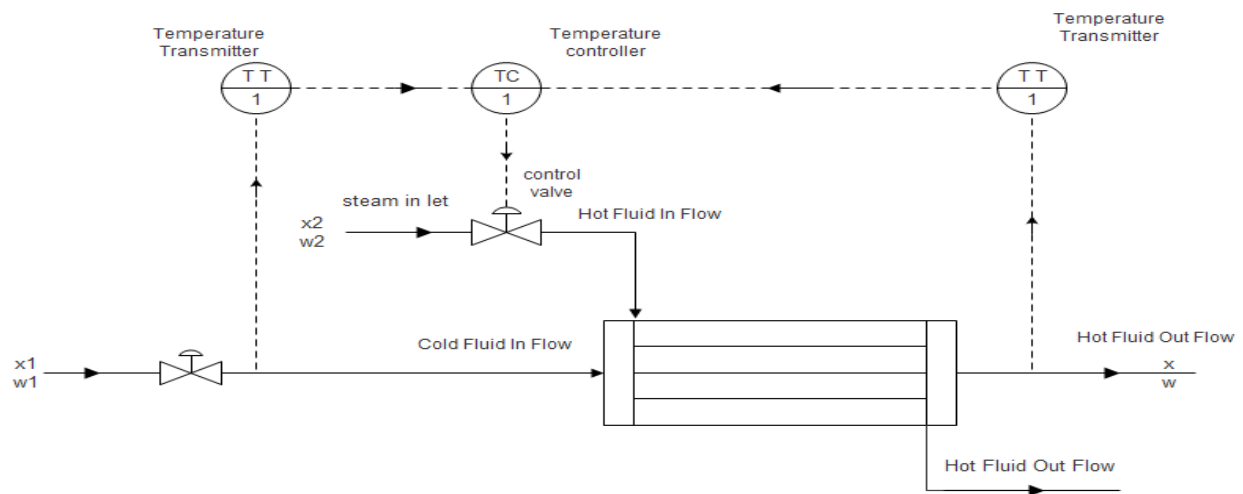


Figure: 3.4 A schematic diagram of strategy3 (feedback-feedforward)

3.5 Feedforward controller

In this research the inherent limitation of feedback controller is that the controller acts after the disturbance distorts the required control objective. If frequent disturbances occur then feedback control will not be able to attain the desired steady state. To limit such kind of drawbacks, feed-forward control is used. Feed-forward control limits the deviation caused by the disturbance but the feed-forward control works in one condition that is the disturbance should be measured or estimated. Feed-forward control can't work alone, so it works alongside feedback control shown Figure 3.5 below.

The transfer function of feed-forward controller can be represented as

$$G_{cf}(s) = -G_d(s)/G_p(s) \quad 3.1$$

Here $G_{cf}(s)$ is the transfer function of feed-back-feed forward controller. $G_p(s)$ is the process transfer function and $G_d(s)$ is the disturbance transfer function.

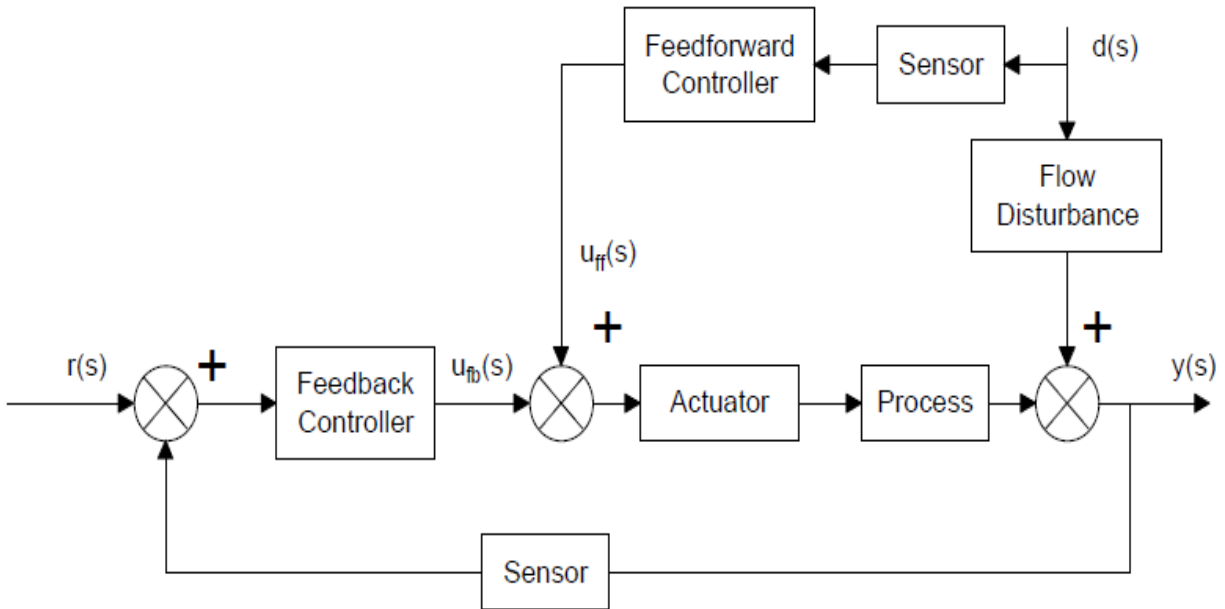


Figure: 3.5 Block diagram of feedback plus feed-forward controller loop

The block diagram of feedback plus feed-forward controller is illustrated in Figure: 3.5. The flow disturbance is measured or estimated and the feed-forward compensator compensates the said disturbance.

The control signal of feedback controller and feed-forward controller is summed up and provided to the process [12].

3.6 Flow chart system

The temperature of the outgoing fluid is measured and compare with a set point, the difference between the actual temperature and set point gives necessary command to the final control element via the actuator unit (valve) .Figure: 3.6 represented flow chart of system

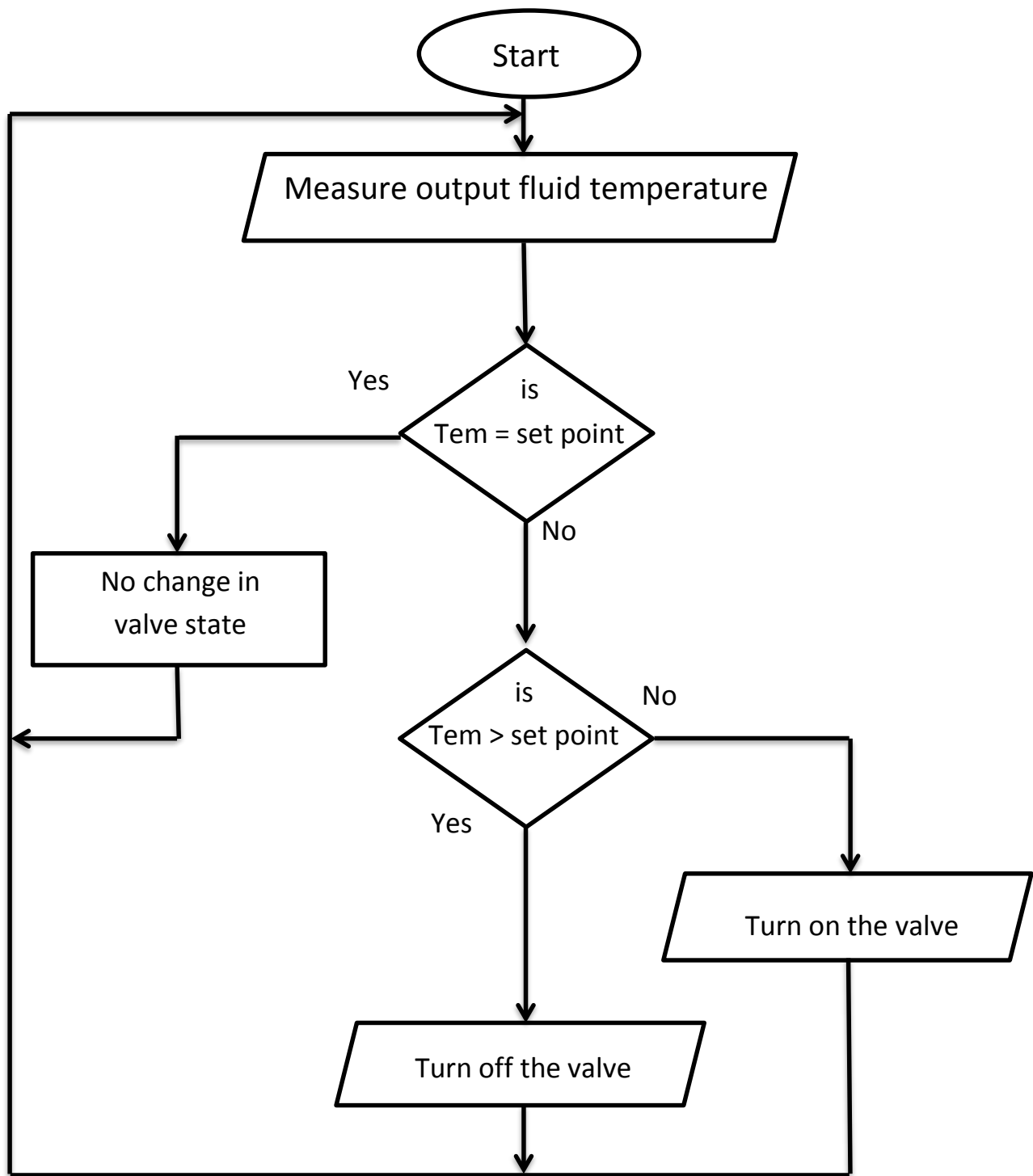


Figure: 3.6 flowchart of the system

Chapter Four

Results & Discussion

4.1 Results

The simulations for the different control mechanism discussed in chapter three were carried out in Simulink and the simulation results have been obtained.

Figure 4.1 and Figure 4.2 represent the Simulink modelling of the heat exchanger system without controller and the system with PID feedback controller.

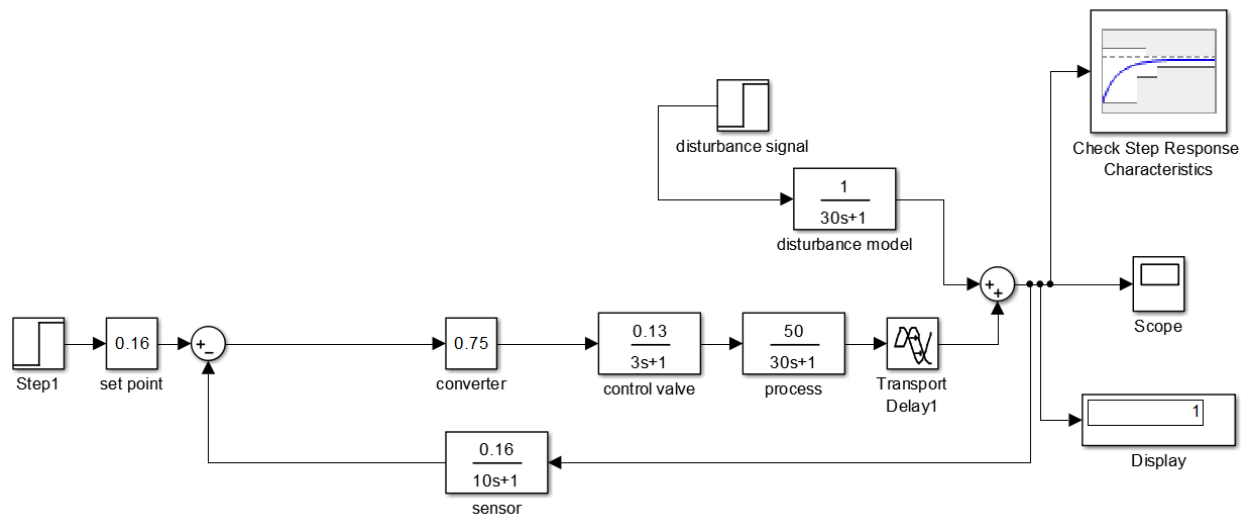


Figure: 4.1 transfer function model of the system without controller

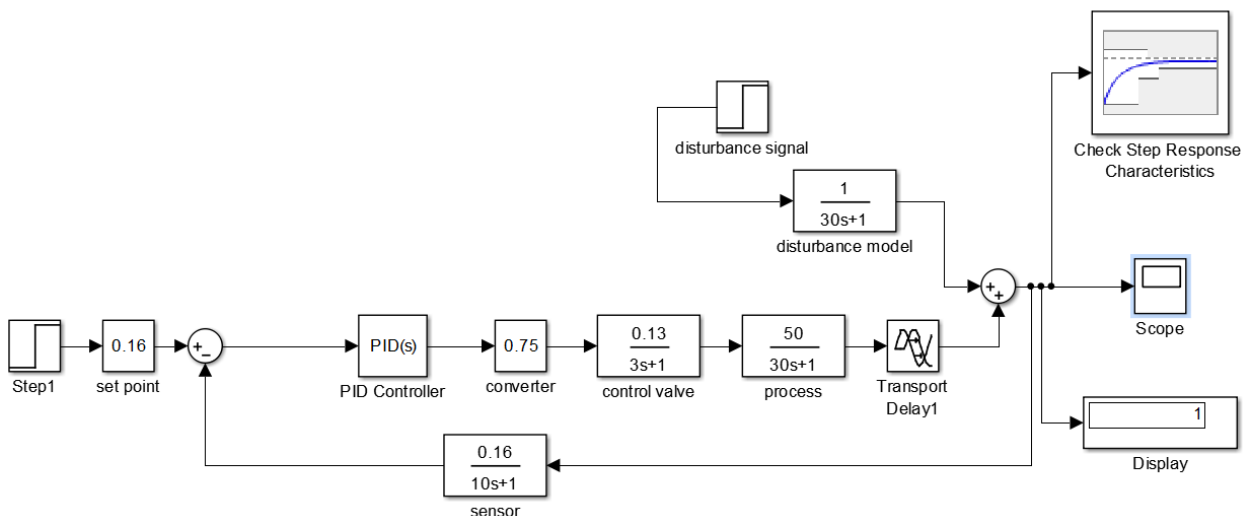


Figure: 4.2 transfer function model of the feedback PID controller using Simulink

For PID controller, the amount of, K_p , K_i and K_d parameters are obtained using matlab tuning method, which is shown in Table 4.1 value of PID parameter.

Table 4.1 value of PID parameter

K_p	K_i	K_d	method
1.0978	0.000026	5.3299	using matlab

The PID controller runs with feed-forward controller in order to reach control purpose. Also analyzing designed controller performance is done in the time domain. Unit step response analysis and performance indicators were used. Processes modeling and controllers' design indicators steps have implemented in MATLAB software Simulink.

The result obtained of system response with feedback controller in Figure: 4.3 and Figure 4.4 below. And Figure 4.5 and 4.6 shown system response without controller then Table 4.2 shows characteristics of the system response with PID and system without controller.

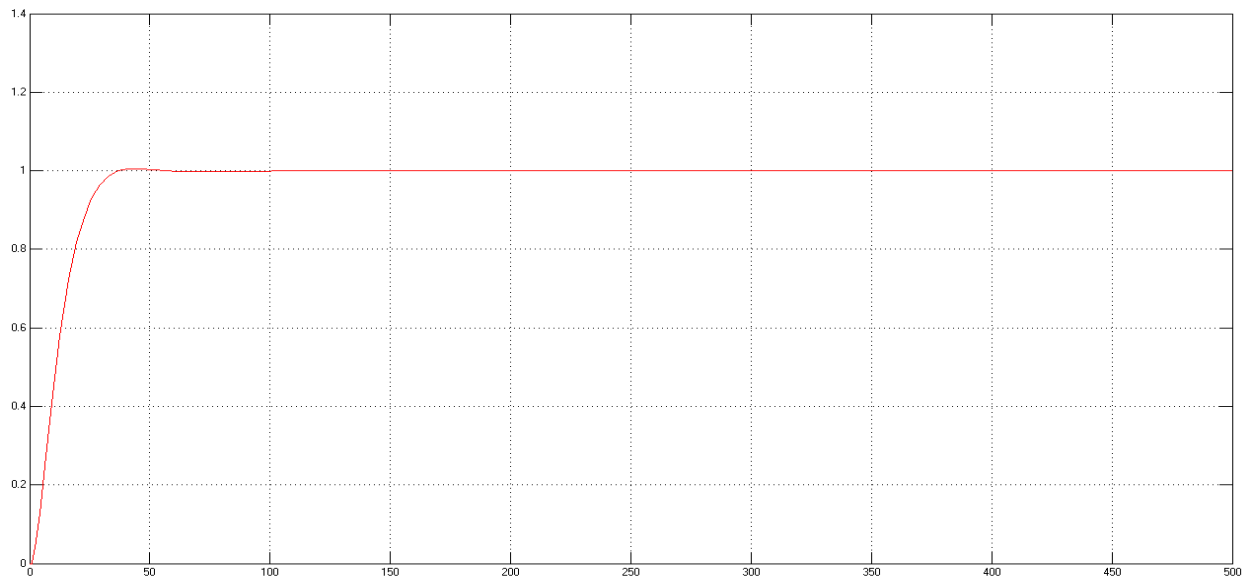


Figure: 4.3 system response with feedback controller (PID)

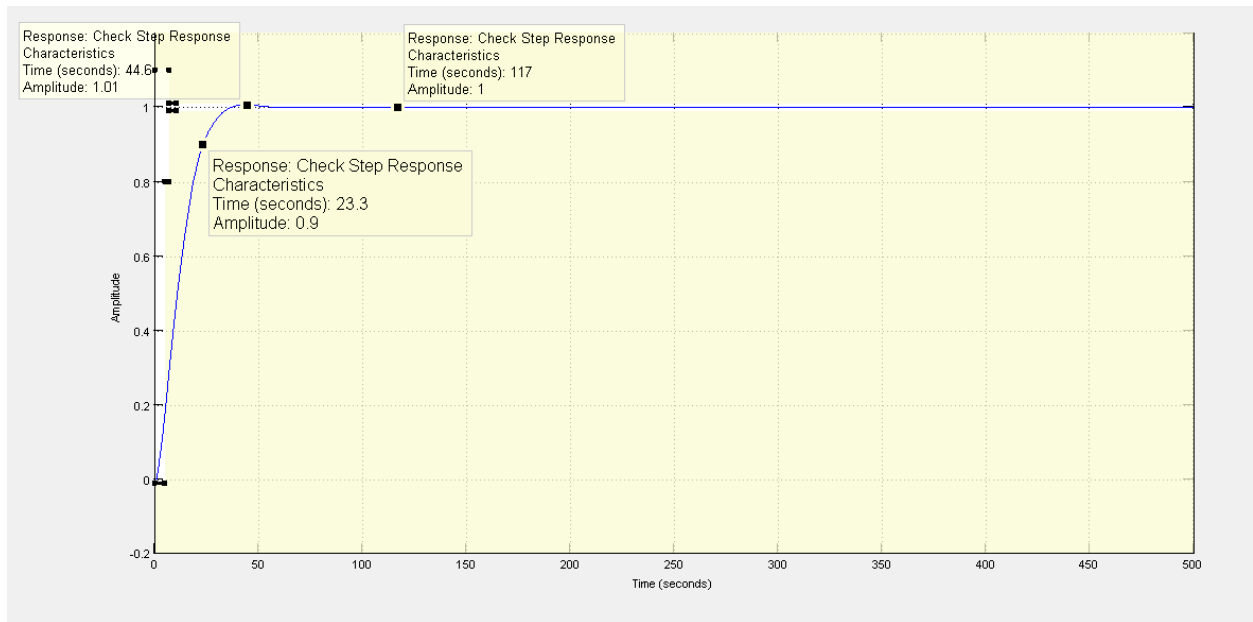


Figure: 4.4 Characteristics of the response system with feedback

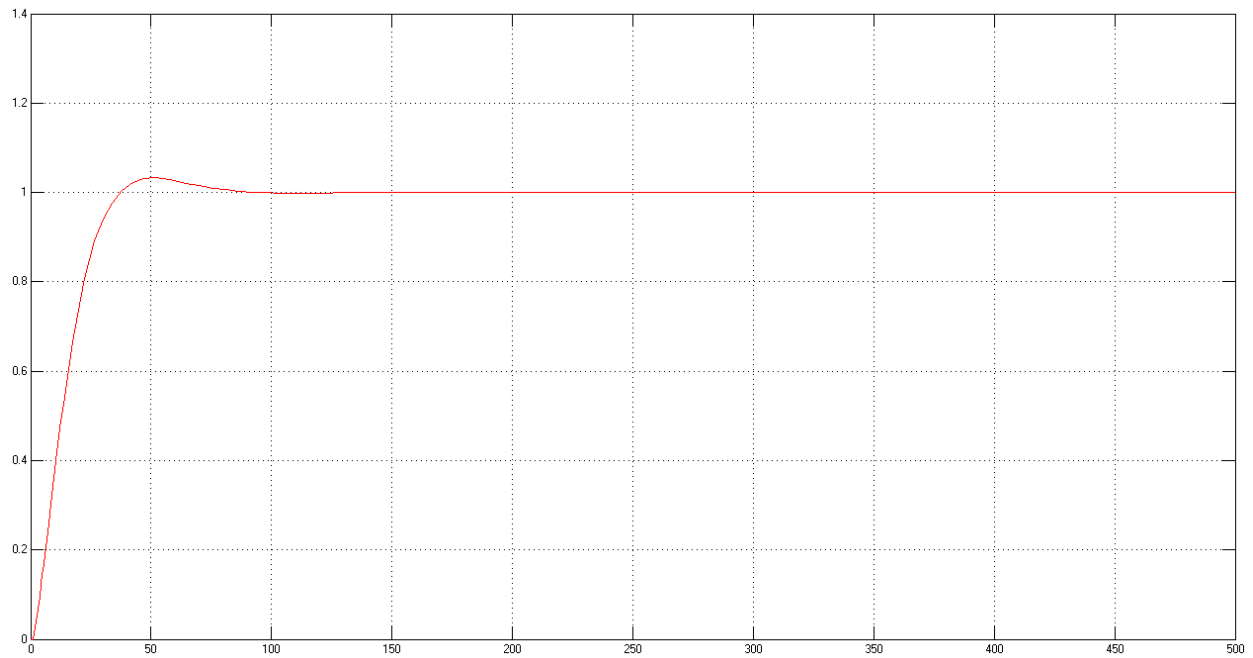


Figure: 4.5 system response without controller

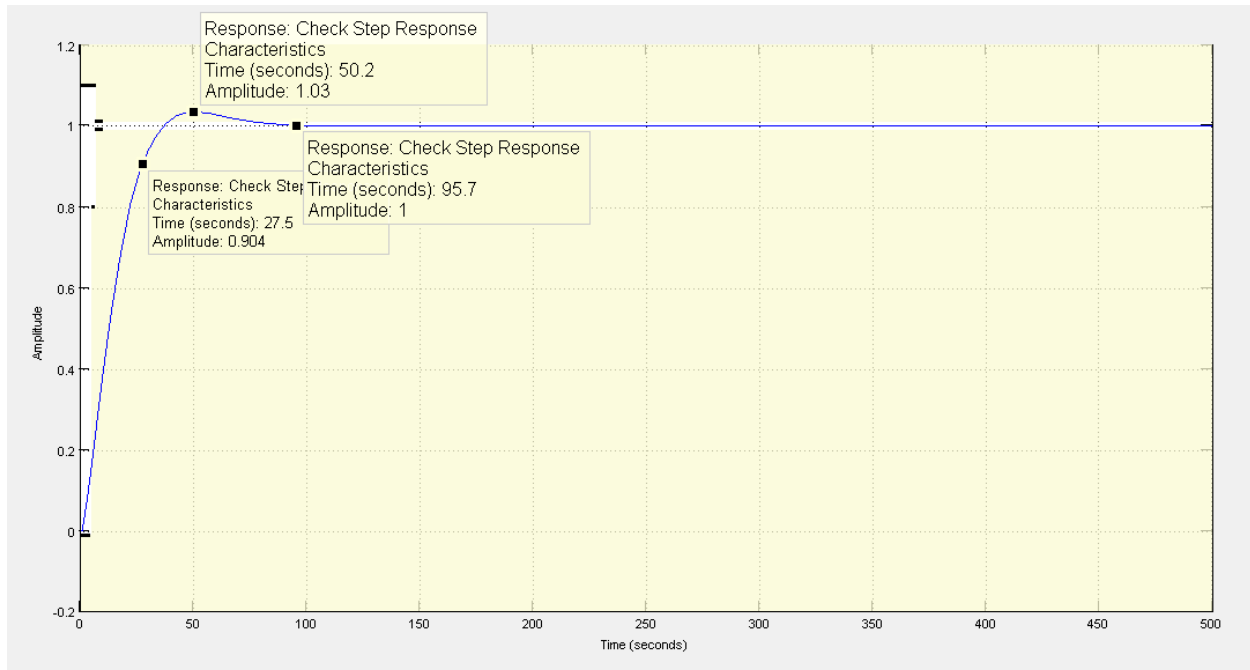


Figure: 4.6 Characteristics of the response system without controller

Table 4.2 Characteristics of the system with feedback

Characteristics	System without controller	System with feedback controller
Settling time	95.7 sec	117 sec
Rising time	27.5	23.3 sec
Over shoot	3%	1.0%
Peak amplitude	1.03	1.01
Final value	1	1

In feedforward control loop, the input fluid flow is measured and the disturbance in the flow is controlled using a feedforward controller. The output of the feedback and the feedforward controller is added and the resultant output is given to the control valve. With the addition of feed forward controller the control performance is further optimized. Figure: 4.7 shows model with feedback-feedforward controller of shell and tube heat exchanger system using MATLAB- SIMULINK.

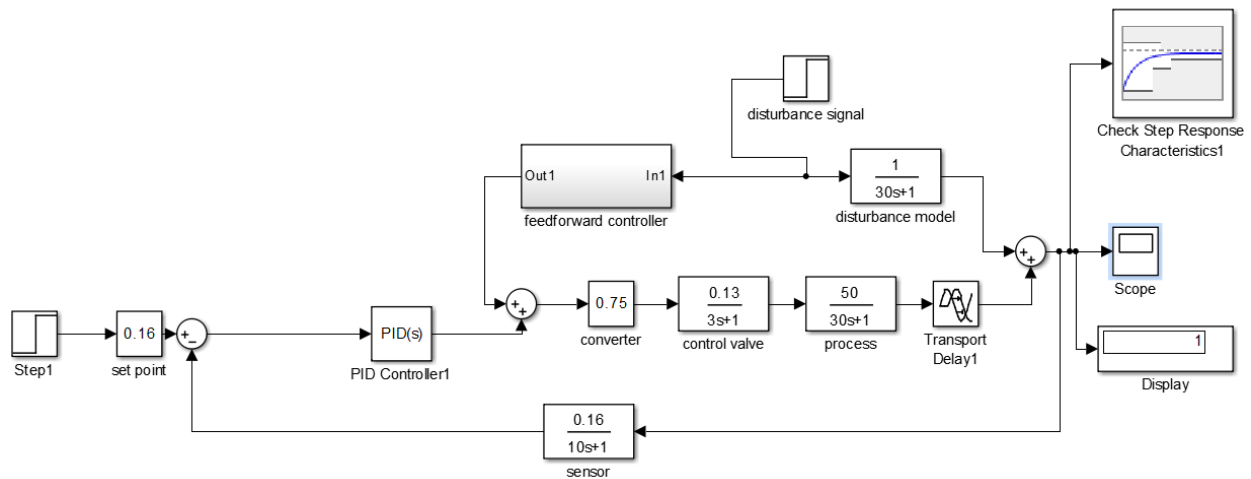


Figure: 4.7 System model with feedback-feedforward controller

In this system after added feed forward and retuning parameter of PID controller the amount of, K_p , K_i and K_d parameters are obtained using matlab tuning method, which is shown in Table 4.3 value of PID parameter.

Table 4.3 value of PID parameter

K_p	K_i	K_D	method
2.2084	0.07585	8.7079	using matlab

The result obtained of system response with feedback- feedforward controllers are shows in Figure 4.8 and Figure 4.9. Then Table 4.4 shows the characteristics of the system response with feedback-feedforward controllers.

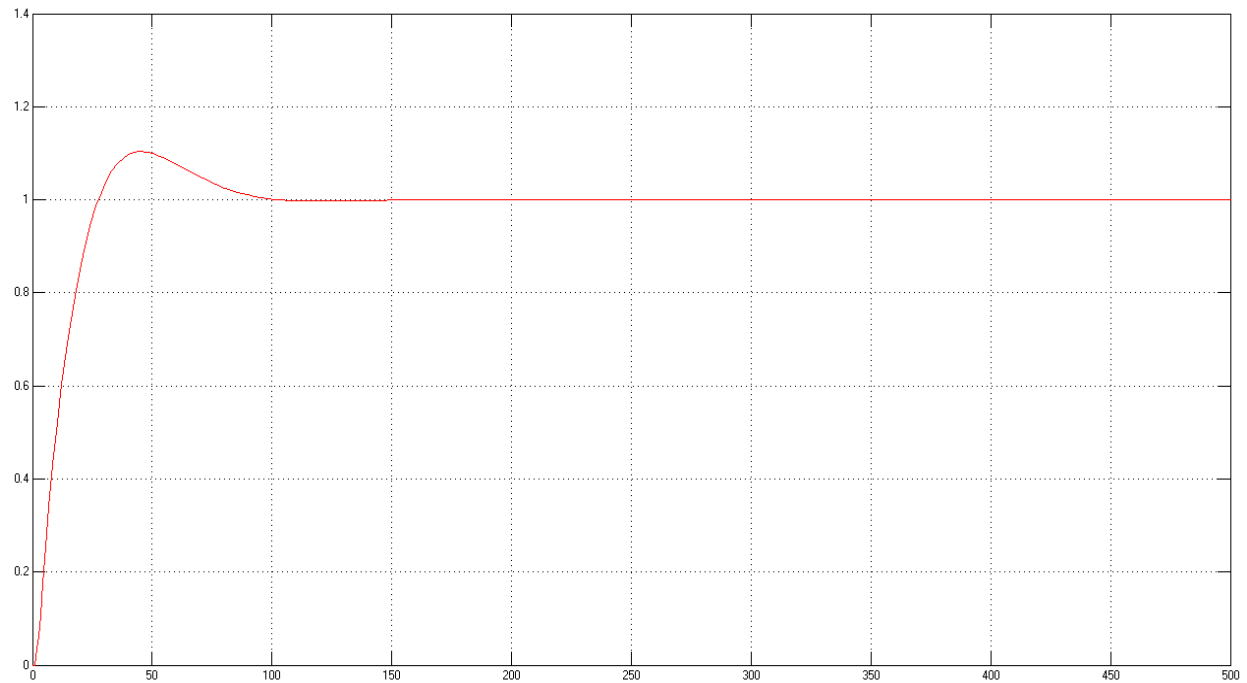


Figure: 4.8 System response with feedback-feedforward controller

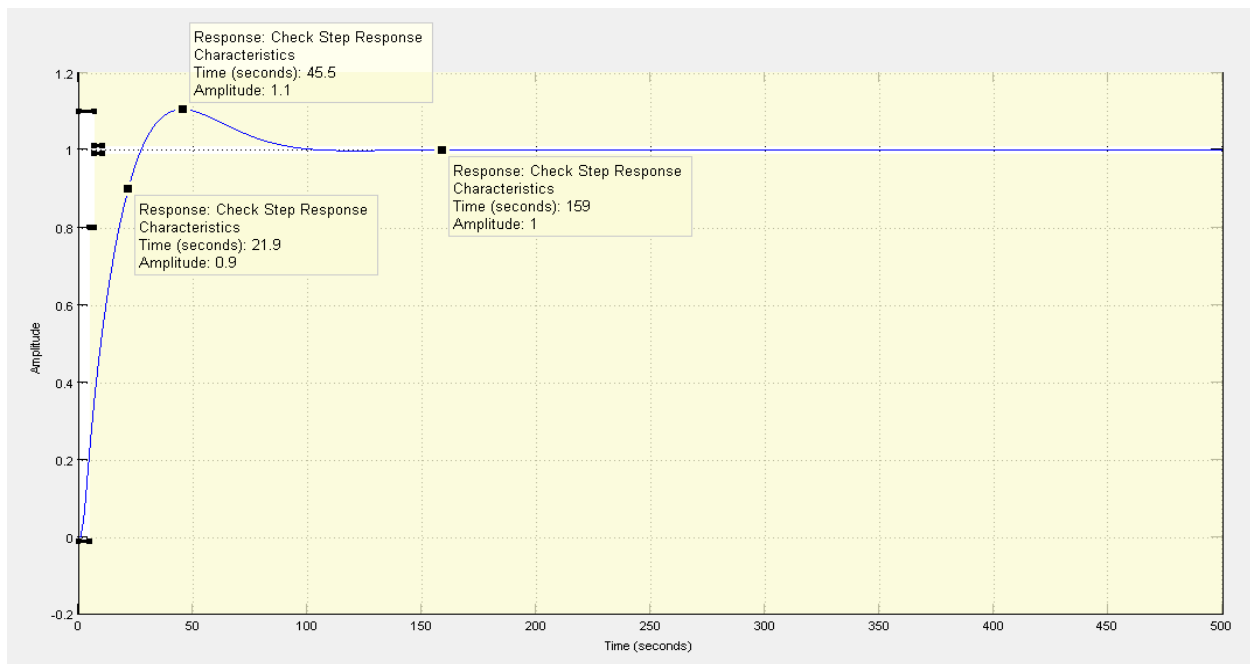


Figure: 4.9 Characteristics of the response system with feedback-feedforward controller

Table 4.4 Characteristics of the system with feedback-feedforward controller

Characteristics	System with fb-ff controller
Settling time	159 sec
Rising time	21.9 sec
Over shoot	10 %
Peak amplitude	1.1
Final value	1

All results obtained are shown in Figure 4.10 below:

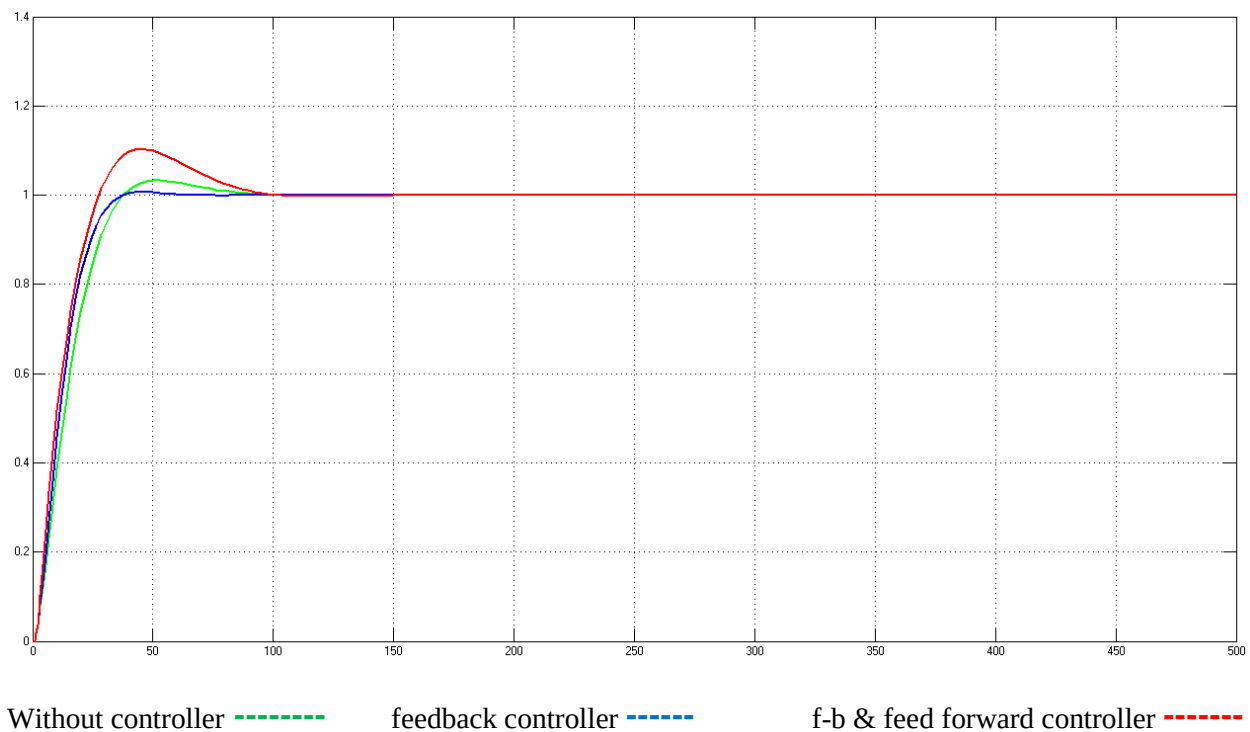


Figure 4.10 response systems without controller & f-b-f-f & feedback controller

4.2 Discussion

In this research designed a feedback PID controller and feedback plus feed-forward controller for controlling output fluid temperature in shell and tube heat exchanger, all results was obtained through research present in Table 4.5 below.

Table 4.5: The results obtained from the systems

controller	S-T	R-T	O-S	P-A	F-V	SS-E
feedback PID	117 sec	23.3 sec	1 %	1.01	1	0
Feedback plus Feed-forward	159 sec	21.9 sec	10 %	1.1	1	0
System without controller	95 sec	27.5 sec	3 %	1.03	1	0

S-T Settling time

O-S Over shoot

F-B Feedback

R-T Rising time

P-A Peak amplitude

F-F Feed Forward

F-V Final value

SS-E steady state error

From the Table 4.5 above the response in the system with feedback controller has very small overshoot 1% compared to the Peak amplitude value of the system but it is not clear shown response in Figure: 4.10. The settling time about 117sec this value of settling time it's acceptable in this system Compared to results obtained from [1] study, the rising time is 23.3 sec also is Acceptable compared to the results obtained from [2] study. Maximum mutation in this control system obtained from feed forward controller it was 10 % with settling time of 159 seconds and rising time 21.9 seconds, this result similar to result was obtained from [3] study but the result in this research by feed forward controller had settling time Slightly lower .

Chapter Four

Conclusion & Recommendations

5.1 Conclusion

In this research, was designed and tuning of PID controller to control the outlet temperature of a shell and tube heat exchanger system. From the simulation results, it is found that the feedback PID controller has a superior performance in the shortest possible time and minimum overshoot irrespective of step change in load and process disturbances than feedback and feedback plus feed-forward controller.

5.2 Recommendations

To enhance response of the system:

- Neuro-fuzzy PID controller can be designed by the implementation of neural network to fuzzy PID controller.
- fuzzy PID controller can be designed surely it will make adaptive tuning values to PID gains

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